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Converting waste PET into dimethyl terephthalate and diverse boronic esters under metal-free conditions†

Minghao Zhang,[a] Yunkai Yu,[a] Zhuo Wang,[a] Shaoyu Zhang,[a] Xiong Gao,[a] Jiaming Liu,[a] Jing Li,[d,e] Weixiang Wu[a,b,c] and Qingqing Mei *[a,b,c]

PET chemical upcycling is essential for advancing sustainable development and a circular economy, while also presenting a dependable option to produce value-added chemicals. Herein, we report a boronic acid involved EG valorization strategy for the upgradation of waste PET into DMT and diverse boronic esters under metal-free conditions without protodeboronation of boronic acids. Based on the remarkable catalytic performance of 1-ethyl-3-methylimidazolium acetate ([EMIm][OAc]) both in PET methanolysis and *p*-tolylboronic acid esterification, this method achieves complete PET degradation, resulting in 99% yield of DMT and 98% yield of 2-(*p*-tolyl)-1,3,2-dioxaborolane (PTDB). This approach not only preserves the high DMT yield in various waste PET and other polyester treatment processes, but also facilitates the transformation of EG into a variety of aryl, heterocyclic, and alkyl boronic esters. The ^{1}H NMR and FT-IR results confirmed that the hydrogen-bonding interaction between [EMIm][OAc] and reactants (PET, EG, and MeOH) enhances both PET methanolysis and boronic acid esterification processes. This method underscores its applicability for upcycling a variety of discarded polyesters and polycarbonates. The conversion of PTDB into other valuable chemicals (phenols, amines, and biaryl compounds) further illustrates the practical utility of this approach in PET disposal.

[a]*Institute of Environment Science and Technology, College of Environmental and Resource Sciences, Zhejiang University, 866 Yuhangtang Road, Hangzhou 310058, Zhejiang, China. E-mail: meiqq@zju.edu.cn*
[b]*Key Laboratory of Environment Remediation and Ecological Health, Ministry of Education, College of Environmental Resource Sciences, Zhejiang University, 310058 Hangzhou, China*
[c]*Innovation Center of Yangtze River Delta, Zhejiang University, Zhejiang 311400, China*
[d]*College of Chemical and Biological Engineering, Zhejiang University, Hangzhou, 310058 Zhejiang, China*
[e]*Institute of Zhejiang University-Quzhou, 99 Zheda Road, Quzhou, 324000 Zhejiang, China*

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Introduction

The improper handling and limited degradability of plastic result in the continued accumulation of waste plastic, posing a significant global environmental threat.[1–5] PET, the most commonly used polyester, is a significant contributor to the growing plastic waste issue.[1,6,7] Currently, PET is mainly recycled through mechanical methods.[6] However, this approach is hindered by a decline in the quality and performance of the recycled plastic, resulting in suboptimal outcomes for the overall recycling strategy.[8–10]

PET chemical solvolysis, serving as a more promising disposal alternative, offers a practical route for valuable chemicals' production.[11–13] Typically, PET solvolysis processes enable the targeted production of TPA/DMT and ethylene glycol (EG) monomers (Fig. 1a).[8,14,15] However, PET upcycling still faces multiple challenges that need to be addressed concurrently, including incomplete depolymerization leading to difficulties in separation and purification, the formation of undesired carboxylate by-products due to metal additives, and limited substrate and product scope. Particularly, the separation energy consumption of EG significantly impacts the production cost of the PET recycling industry due to its high boiling point of 197.6 °C, its considerable viscosity, and its high solubility in water.[11,16] Recent advancements in targeted EG upgradation strategies have facilitated the development of designed conversion patterns aiming at circumventing issues related to low value and separation. Based on oxidation, reduction, or EG capturing strategies, various value-added chemicals, such as carboxylic acids or carboxylates,[12,13,17–20] silicon- or boron-containing chemicals,[11,21–23] and dimethyl carbonate,[16] have been proposed for EG upgradation with attractive achievements (Fig. 1b). The development of facile and low-carbon methodologies remains an area ripe for innovation.

Boronic esters represent a significant class of organoboron compounds, recognized for their exceptional material properties and diverse synthetic applications.[24–26] Numerous studies have employed boronic esters as feedstocks to develop

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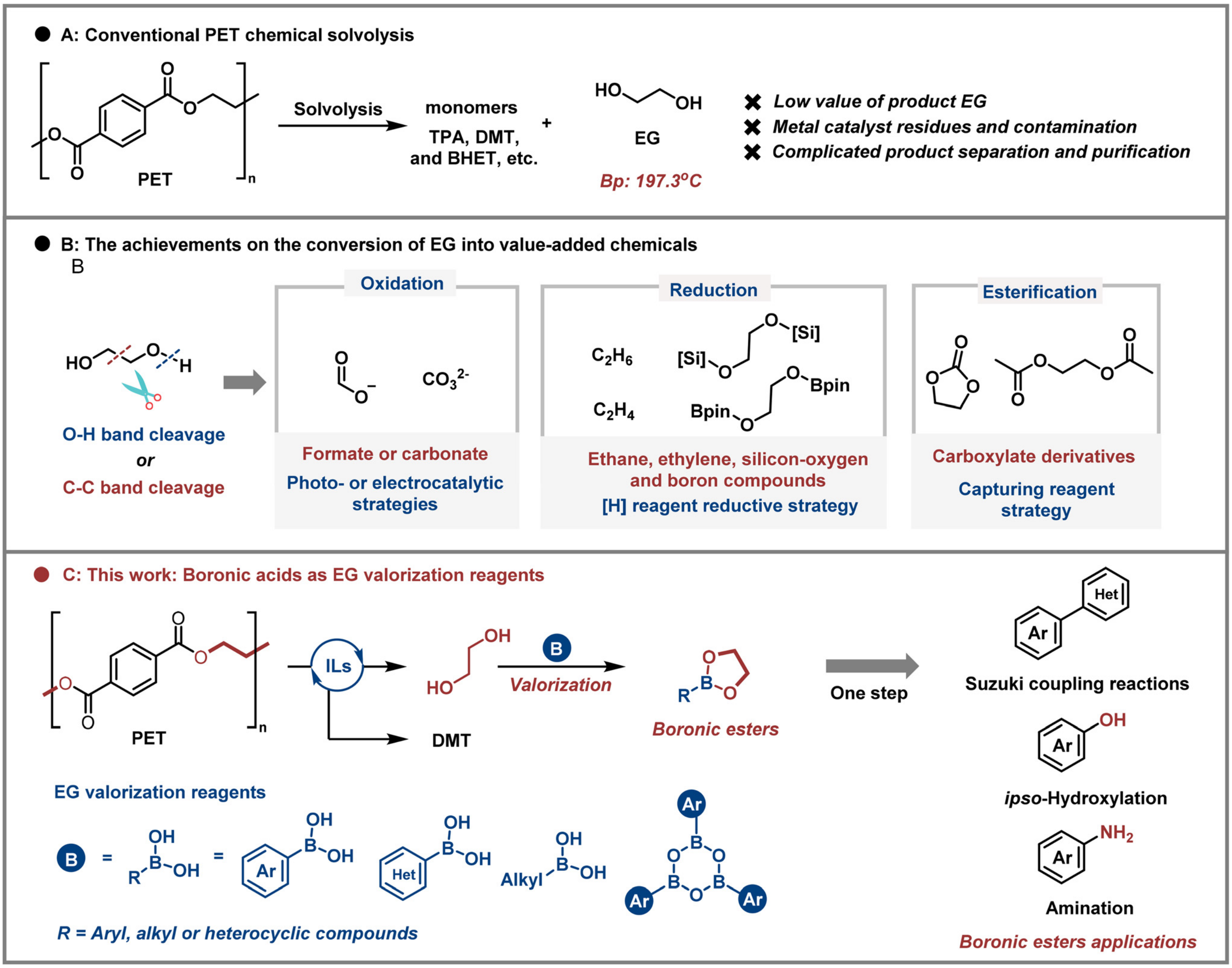

Fig. 1 IL-catalyzed EG valorization strategy for complete upgradation of PET to DMT and boronic esters.

multifunctional new materials and value-added chemicals, including amines, phenols, and aromatic compounds.[25–29] Typically, boronic esters are synthesized through the reversible binding of diols and boronic acids. Given the low value of EG and the potential for comprehensive utilization of the carbon skeleton of EG units, the conversion of EG into boronic esters emerges as a promising strategy. Our recent research has demonstrated that utilizing boronic acids to *in situ* capture EG and convert it into boronic esters is an effective method for achieving complete PET methanolysis and EG upgradation.[11] However, during the layered double oxide (LDO)-catalyzed process, the formation of the by-product toluene (4–9%) through the protodeboronation of boronic acids can significantly reduce the selectivity of EG valorization.[11] Furthermore, since LDO is a heterogeneous catalyst, the crystallized DMT becomes intermixed with the catalyst after the reaction, thereby complicating the separation process of DMT. Therefore, it is essential to develop a practical homogeneous PET–boronic acid upcycling system to facilitate the efficient separation of DMT, while simultaneously suppressing the undesired protodeboronation of boronic acids to enhance the selectivity of boronic esters.

Ionic liquids, leveraging their synergistic anionic and cationic catalytic attributes, have been effectively deployed as catalysts in the realm of metal-free polyester alcoholysis.[30–34] For example, [BMIm][OAc]-catalyzed methanolysis was proposed for converting waste PEF into dimethyl furan-2,5-dicarboxylate (DMFD) with 77.6% yield at 130 °C for 30 minutes.[32] This approach also facilitated the conversion of PET to DMT, yielding 41.7% at 150 °C. Heterogeneously, PIL–Zn^{2+} exhibited efficient catalytic performance in the methanolysis of PET at 170 °C, resulting in a DMT yield of 90.3%.[35] A comprehensive review of the literature pertaining to IL-catalyzed PET methanolysis is provided in Table S1 (refer to the ESI† for details). Although excessive ionic liquid usage and incomplete depolymerization remain critical challenges in ionic liquid-catalyzed polyester methanolysis, the tunable properties of ionic liquids present a promising opportunity to address these issues within the PET–boronic acid upcycling system.

Herein, we proposed an ionic liquid (IL) catalyzed *ex situ* EG capturing approach to completely inhibit the protodeboronation of boronic acids while maintaining a high yield of DMT. The catalytic efficacy of [EMIm][OAc] in activating carbonyl and alcohol hydroxyl groups resulted in successful *ex situ* EG valorization. By introducing *p*-tolylboronic acid (PTBA) into the post-PET methanolysis system at room temperature, DMT and 2-(*p*-tolyl)-1,3,2-dioxaborolane (PTDB, **3a**) were produced with high yields of 99% and 98%, respectively. Notably, no protodeboronation by-product (toluene) was detected. This approach not only shows broad commercial polyester disposal scope, but also proves effective for converting EG into aromatic or alkyl boronic esters. The application experiments of the obtained PTDB were expanded to synthesize other valuable chemicals (phenols, amines, and biaryl compounds), thereby confirming the practicality of this EG valorization strategy and offering new insights for PET disposal.

Results and discussion

To ensure product purity and prevent metal contamination, as well as avoid the formation of unwanted carboxylate by-products due to metal catalysts, ILs were selected as the preferred catalysts for the PET valorization process.[33,36–40] After screening the reaction conditions, as shown in Table S2,† the optimal *in situ* PET upcycling reaction conditions were as follows: PET (1 mmol), PTBA (1 mmol), [EMIm][OAc] (5 mol%) and MeOH (3 mL) at 180 °C for 1.5 hours (Table S2 and Fig. S1b†). Under the optimal conditions, PET was completely converted *in situ*, providing 88% PTDB and 99% DMT, respectively. The protocol was also found to be suitable for disposing of diverse commercial waste PET or other polyesters to produce DMT or other carboxylic acid esters, respectively (Fig. S4†). However, undesired levels of protodeboronation of PTBA to toluene (12–45%) were observed during both reaction condition optimization and substrate scope expansion, leading to decreased yields of PTDB (Table S2 and Fig. S4†). Although controlling the catalyst basicity strength and the amount of base can partially inhibit protodeboronation, it cannot eliminate this undesirable side reaction. Given the significant value and practical applications of boronic esters, there is a pressing need to devise a highly effective method to combat the challenge of protodeboronation of boronic acids.

The production of toluene is mainly due to the significant protodeboronation activity of basic catalysts at elevated temperatures, which enables the rapid protodeboronation of boronic acids in the presence of water molecules. Lowering the temperature of the system could reduce the formation of undesired toluene, but it could also result in incomplete depolymerization. Based on the excellent catalytic activity of [EMIm][OAc] for sole PET methanolysis (Fig. S1c†) and avoiding temperature-affected protodeboronation of boronic acids, an EG *ex situ* capturing approach was proposed. This modified approach involves sole PET methanolysis catalyzed by [EMIm][OAc] at 180 °C followed by EG esterification with boronic acids at room temperature. The modified approach was effective, with negligible occurrence of protodeboronation of arylboronic acids and a PTDB yield of up to 98% within 15 minutes, as described in Fig. S1d and S2.† In the treatment of commercial waste PET or other polyesters using this modified approach, the addition of *p*-tolylboronic acid post-PET degradation significantly mitigates toluene by-product production and notably enhances the yield of **3a** (Fig. S2†). This approach enabled the effective degradation of various PET feedstocks, such as fabrics, transparent films, woven mesh, woven tape, colored pallets, and different bottles, resulting in 78–97% yields of **3a** and 84–99% yields of DMT (Fig. 2 and Fig. S5†). Commercial polyesters such as poly(ethylene succinate) (PES), polyethylene 2,5-furandicarboxylate (PEF), and poly(ethylene adipate) (PEA), as well as polycarbonates like poly(propylene carbonate) (PPC), can also be recycled and converted into various value-added dicarboxylates in 96–98% yield and (*S*)-4-methyl-2-(*p*-tolyl)-1,3,2-dioxaborolane in 92% yield.

For targeted conversion of EG to diverse boronic esters, the scope of boronic acids was also analysed using this modified approach, as depicted in Fig. 3. The results demonstrate that the reaction exhibits excellent functional group tolerance with negligible impact on the esterification reaction. Specifically, phenylboronic acids and arylboronic acids with electron-donating groups (*p*-*iso*-Pr, *p*-*t*-Bu and *p*-OMe) on the benzene ring exhibited outstanding selectivity, leading to the formation of **3b–3e** in yields ranging from 93% to 98%. Additionally, *p*-hydroxyphenylboronic acid, containing highly reactive phenolic hydroxyl groups, was readily converted to 4-(1,3,2-dioxaborolan-2-yl)phenol (**3f**) in 92% yield. Furthermore, modifiable halogen groups (*p*-F, *p*-Cl, and *p*-Br) and strong electron-withdrawing groups (*p*-CO_2Me, *p*-CF_3, *p*-$COCH_3$, and *p*-NO_2) were well tolerated, delivering products **3g–3m** in 92–98% yield. Arylboronic acids bearing functional groups at the *meta*-position of the benzene ring were also detected, yielding **3m–3o** in 95–98% yield. This underscores the minimal impact of steric hindrance on the current approach. Subsequently, boronic acids containing fused aromatic (naphthalene and phenanthrene), heterocyclic (thiophene), heteroatom-linked aromatic (triphenylamine) or cycloalkyl ring (cyclohexane) structures were also displayed to be compatible with this modified approach and converted to target boronic esters **3q–3v** in 83–95% yield. Additionally, diboronic acids, including 1,4-phenylenediboronic acid and tetrahydroxydiboron, were also proved as efficient EG upgradation reagents, providing the target products **3w** and **3x** with yields of 93% and 65%, respectively. 2,4,6-Triphenylboroxine, typically derived from the trimerization of boronic acids and possessing more available reaction sites, represents a promising candidate for potential EG trapping reagents. Under standard reaction conditions, the triarylboroxines **2b** and **2c** smoothly converted to the target products **3b** and **3g**, yielding 91% and 94%, respectively. Furthermore, gram-scale experiments of PET with *p*-tolylboronic acid were also conducted as described in Fig. 3, confirming the potential application value of the modified PET upcycling approach.

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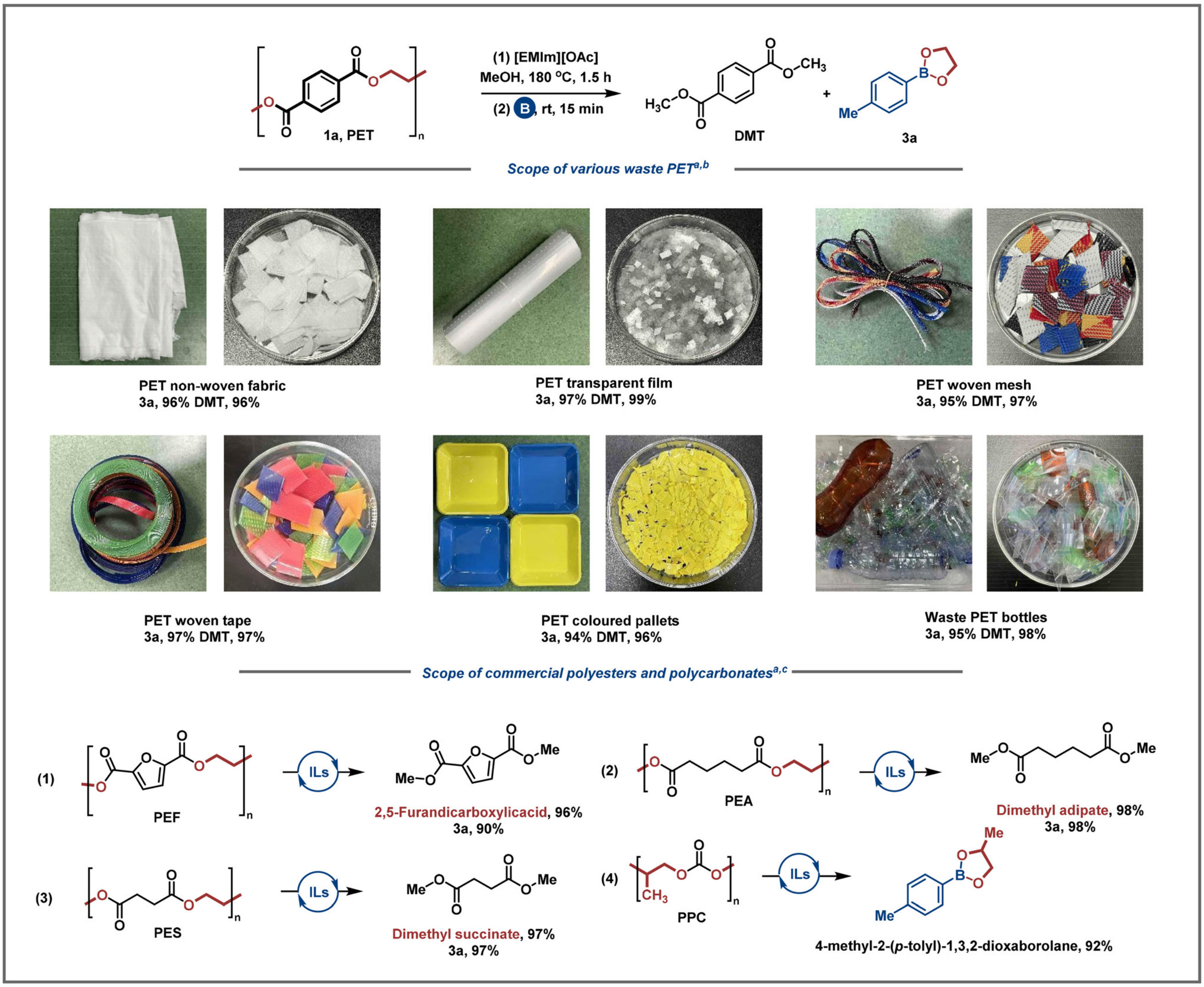

Fig. 2 Scope of polyesters. [a] Standard reaction conditions: (1) polyester (1 mmol), [EMIm][OAc] (5 mol%), and MeOH (3 mL) at 180 °C for 1.5 h. (2) Boronic acids (1 mmol) at room temperature for 15 min. [b] Yields were determined by GC analyses with mesitylene as the internal standard after solvent removal by rotary evaporation. [c] Yields were determined by ^{1}H NMR analyses with mesitylene as the internal standard after solvent removal by rotary evaporation.

To further demonstrate the effectiveness of this approach, experiments were conducted to compare the reactivity of arylboronic acid **2a** and arylboronic ester **3a** (Fig. 3, entries 3 and 4). In the Suzuki–Miyaura cross-coupling reaction between organoboron compounds and pyridine-based heterocyclic compounds, it was observed that arylboronic esters **3a** demonstrated superior reactivity and selectivity compared to *p*-tolylboronic acid, reacting with 4-iodopyridine to provide 4-(*p*-tolyl)pyridine in 86% yield.[28] The conversion of arylboronic esters into other valuable chemicals was also investigated, emphasizing the versatility of the boronic acid ester functional group, which can act as a synthon for functional group transformations (Fig. 3, entries 5 and 6). Furthermore, the transformation of arylboronic ester **3a** into *p*-cresol or *p*-toluidine was conducted, yielding 93% and 62%, respectively, through *ipso*-hydroxylation or amination processes.[26,29] In summary, these extensive reactions demonstrate broad scope of substrate adaptability, coupled with the various applications of the boronic ester product, emphasizing the potential practical value of this modified approach.

Research studies and experimental results indicate that the formation of hydrogen bonds between [EMIm][OAc] and reactants plays a crucial role in initiating the cleavage of the ester bond in PET.[32,35,40,41] Fig. 4a illustrated the potential hydrogen bonding and self-cyclization of PTBA catalyzed by [EMIm][OAc]. To validate these hypotheses, ^{1}H NMR analysis was conducted to study the interaction between the [EMIm][OAc] and reactants with varying molar ratios, as illustrated in Fig. 4b–f. The C2-hydrogen of the [EMIm][OAc] shifted towards the high field from 9.04 ppm to 8.84 ppm with the molar ratio of methyl benzoate (MB) to [EMIm][OAc] increased from 0 : 1 to 5 : 1. This result indicated the formation

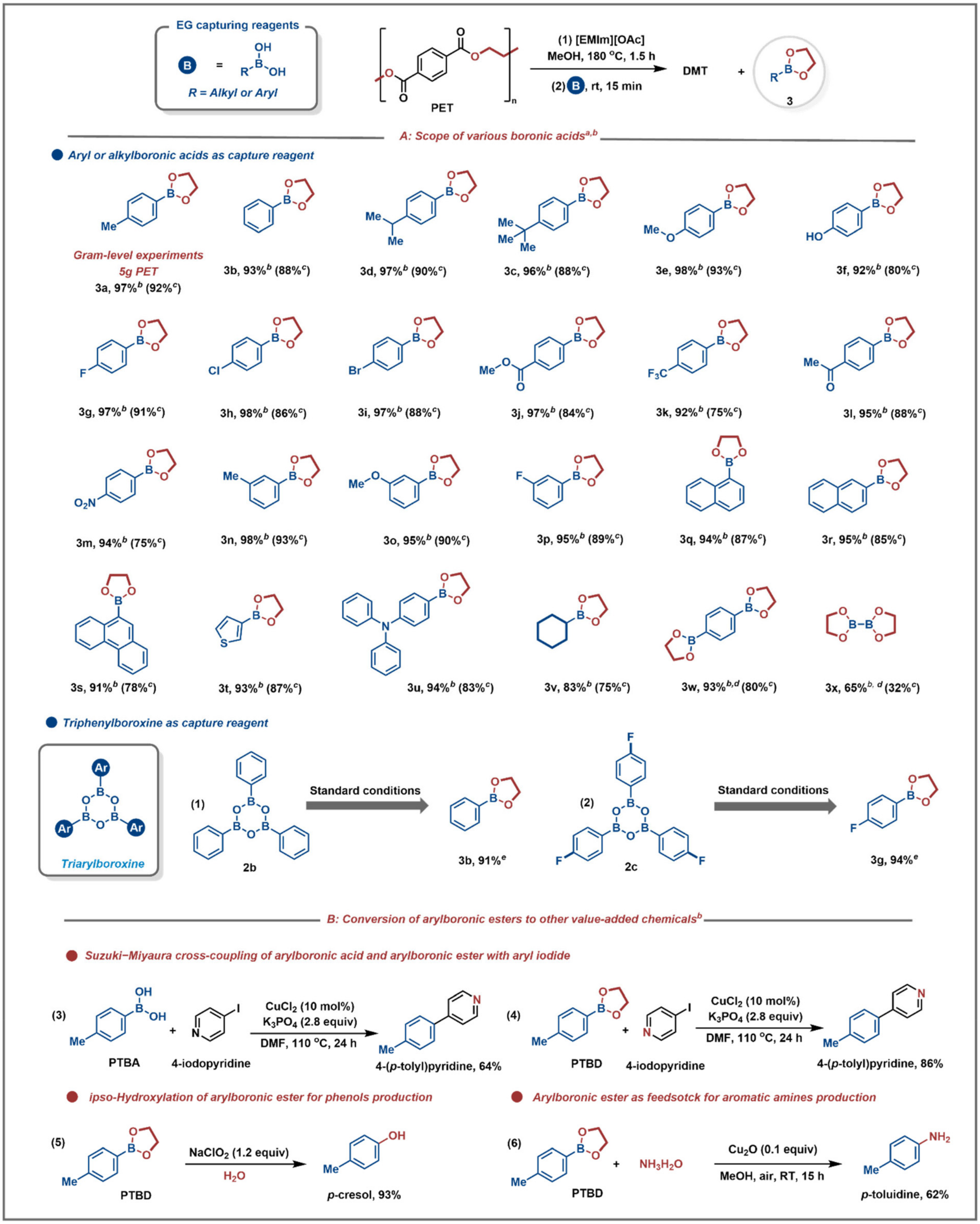

Fig. 3 Scope of boronic acids and boronic ester applications. [a] Standard reaction conditions: (1) PET (1 mmol), [EMIm][OAc] (5 mol%), and MeOH (3 mL) at 180 °C for 1.5 h. (2) Boronic acids (1 mmol) at room temperature for 15 min. [b] Yields were determined by ^{1}H NMR analyses with mesitylene as the internal standard after solvent removal by rotary evaporation. [c] Isolated yields. [d] Boronic acids (0.45 mmol), [e] boronic acids (0.3 mmol).

of hydrogen bonds between the carbonyl oxygen of methyl benzoate and $[EMIm]^+$, which enhances its electrophilic properties and further activates the carbonyl carbon.[32] The chemical shift of the hydroxyl hydrogen was observed to reach 0.77 ppm downfield when the molar ratio of MeOH to [EMIm][OAc] was 1 : 0 and 1 : 1. Increasing the concentration

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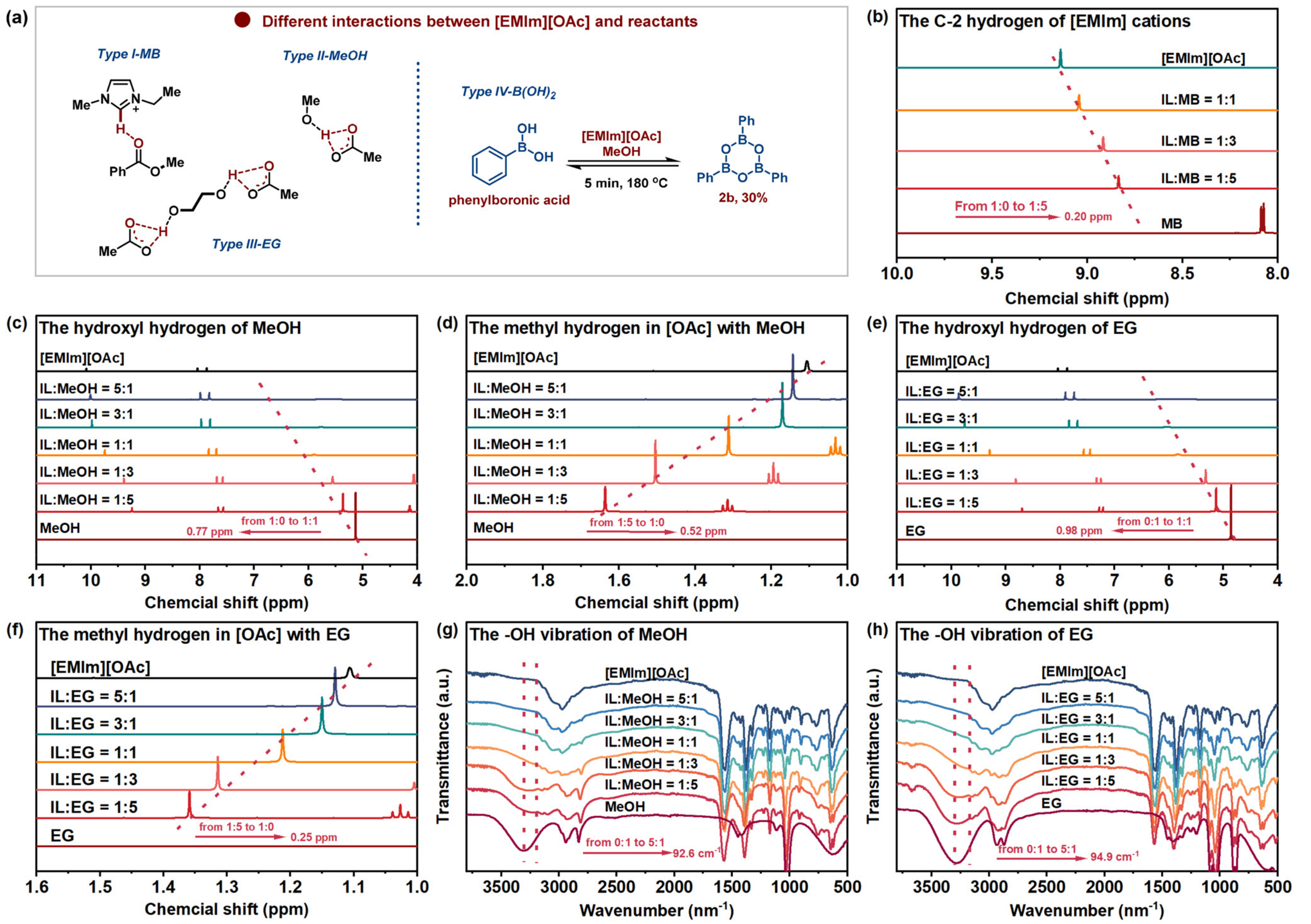

Fig. 4 Characterization of different hydrogen bonds formed during PET depolymerization. (a) Different hydrogen bond interactions between [EMIm][OAc] and reactants. (b–f) ^{1}H NMR spectra of the [EMIm][OAc] and different reactants mixture with different molar ratios. ^{1}H NMR spectra were recorded at room temperature in DMSO-d_6. (g and h) FT-IR spectra of [EMIm][OAc] and alcohol mixtures of different molar ratios.

of [EMIm][OAc] further led to the broadening and gradual disappearance of the hydroxyl hydrogen signal, confirming the formation of a strong hydrogen bond between the hydroxyl hydrogen of MeOH and $[OAc]^-$. Consequently, this hydrogen bond enhances the nucleophilicity of MeOH, promoting the nucleophilic attack on the carbonyl carbon of PET and facilitating efficient C–O bond cleavage. Additionally, the methyl hydrogen shifts of $[OAc]^-$ reached 0.52 ppm as the molar ratio of [EMIm][OAc] to MeOH increased from 1 : 5 to 1 : 0 indicating effective electron transfer of hydrogen bonds. Furthermore, the interaction between EG and [EMIm][OAc] was examined, confirming the similar chemical shift of hydroxyl hydrogen of EG and the methyl group of $[OAc]^-$. Specifically, the hydroxyl hydrogen of EG shifted 0.98 ppm downfield, while the methyl group shifted 0.25 ppm upfield as the molar ratio of [EMIm][OAc] to EG increased from 1 : 5 to 1 : 0. This observation validates that the [EMIm][OAc] boosts the nucleophilicity of EG, facilitating the esterification reaction of EG with boronic acids.[31] Additionally, FT-IR analysis further confirmed a significant interaction between compound [EMIm][OAc] and alcohols reactants (MeOH and EG), as illustrated in Fig. 4g and h. Noticeable redshifts in the hydroxyl vibration peak were observed. Specifically, both the hydroxyl vibration peaks of MeOH and EG became broader and shifted from 3316.3 to 3223.7 cm^{-1} and 3293.1 to 3198.2 cm^{-1}, respectively, as the molar ratio of [EMIm][OAc] to MeOH increased from 0 : 1 to 5 : 1. These observations provide further reliable evidence supporting the formation of robust hydrogen bonds between $[OAc]^-$ and hydroxyl hydrogens of MeOH or EG.[32] In summary, the enhanced reactivity of the reactants, facilitated by the formation of individual hydrogen bonds with the anions or cations of [EMIm][OAc], resulting in the efficient cleavage of the ester bond in PET methanolysis and boronic acid esterification with EG.

A possible boronic acid involved PET upgradation mechanism was proposed based on our previous research and these observed interactions of hydrogen bonds between [EMIm][OAc] and reactants (Fig. 5).[11,12,16,32] Initially, the hydrogen bond formed between $[EMIm]^+$ and carbonyl oxygen of PET led to an enhanced electropositive nature of the carbonyl carbon. This activation leads to a more favorable nucleophilic attack from activated MeOH by forming a hydrogen

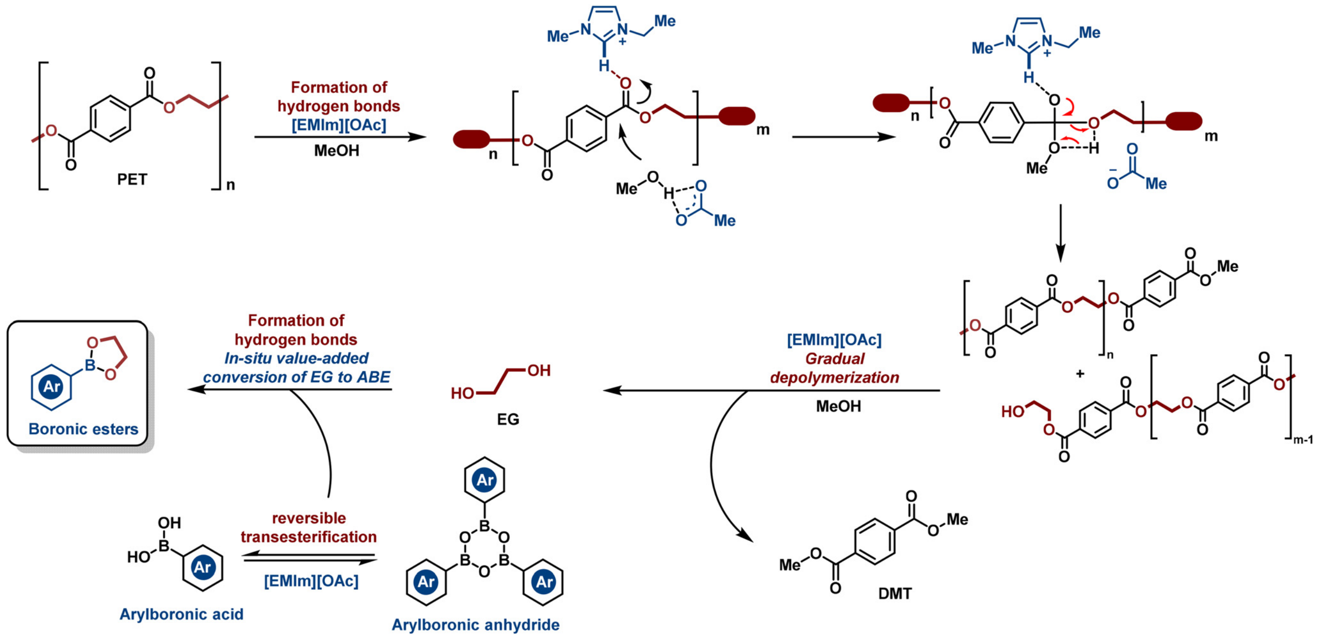

Fig. 5 The possible [EMIm][OAc]-catalyzed PET methanolysis coupled with *in situ* EG esterification with boronic acids.

bond with $[OAc]^-$, resulting in the formation of a tetrahedral intermediate. Electron transfer within this intermediate facilitated the cleavage of the C–O bond, generating DMT, EG, and oligomers as the depolymerization proceeded.[32] The *in situ* produced EG was activated by the hydrogen bond and simultaneously captured by boronic acid, leading to the formation of boronic esters. The *in situ* capture of EG with boronic acids biased the reaction equilibrium towards increased DMT and EG production, leading to enhanced depolymerization of oligomers and complete PET upgradation. These findings confirm that the activation effect of [EMIm][OAc] in combination with reactants improves both methanolysis and EG upgradation with boronic acid, resulting in a more efficient *in situ* conversion process.

In this study, we proposed a boronic acid involved PET degradation strategy for the *in situ* or *ex situ* upgradation of waste PET to DMT and boronic esters under metal-free conditions. In the *in situ* reaction process, [EMIm][OAc] displayed remarkable catalytic performance both in PET methanolysis and boronic acid esterification, producing DMT and 2-(*p*-tolyl)-1,3,2-dioxaborolane (PTDB, **3a**) in 99% and 88% yield, respectively. The improved method addressed the protodeboronation of boronic acids and enhanced the yield of **3a** up to 98%. Aromatic, fused aromatic, heterocyclic, heteroatom-linked aromatic, or alkyl boronic acids were applied for EG upgradation to produce diverse boronic esters and reached 83%–98% yields at room temperature within 15 minutes. The produced PTDB was utilized for the synthesis of other valuable chemicals (phenols, amines, and biaryl compounds) in moderate to excellent yields after the product separation process. ^{1}H NMR and FT-IR analyses confirmed the pivotal role of the multiple hydrogen bonds formed between reactants (PET, MeOH, and EG) and [EMIm][OAc], promoting the PET methanolysis and boronic acid esterification. This modified approach also enables various PET-based, polyesters and polycarbonates upgradation to carboxylic and boronic esters, delivering a novel and practical option for waste plastic treatment.

Author contributions

Minghao Zhang: experiments, formal analysis, and writing – original draft. Yunkai Yu: formal analysis. Zhuo Wang: formal analysis. Shaoyu Zhang: formal analysis. Xiong Gao: experiments and NMR analysis. Jiaming Liu: formal analysis. Jing Li: NMR analysis, Weixiang Wu: formal analysis and resources. Qingqing Mei: conceptualization, formal analysis, funding acquisition, resources, supervision and writing.

Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

The authors declare no competing interests.

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